

* Maps-:

Maps are a data structure that provides key-value mapping.

- The zero value of a map is nil.

```
ages := make(map[string]int)
ages["John"] = 37
ages["Mary"] = 24
```

- The len() function works on a map. It returns the total number of key/value pairs.

* Mutations-:

- Insert an Element-:

```
m[key] = elem
```

- Get an Element-:

```
elem = m[key]
```

- Delete an Element-:

```
delete(m, key)
```

- check if a key exists-:

```
elem, ok := m[key]
```

- If key is in m, then ok is true and elem is the value as expected.
- If key is not in the map, then ok is false and elem is the zero value of the map's element type.

* Key Types-:

Map keys may be of any type that is comparable. The language spec defines this precisely, but in short, comparable types are boolean, numeric, string, pointer, channel, and interface types, and structs or arrays that contain only those types.

- Notably absent from the list are slices, maps and funcs.
- These types cannot be composed using `==`, and may not be as map keys.

```
bits := make(map[string]map[string]int)
```

* Count Instances-:

You can combine an if statement with an assignment operation to use the variable inside the if block:

```
if _, ok := names["Dennis"]; !ok {
    ...
}
```

* Effective Go-:

- Like slices, Maps hold references-:

Like slices, maps hold references to an underlying data structure. If you pass a map to a function that changes the contents of the map, the changes will be visible in the caller.

- Map Literals-:

Maps can be constructed using the usual composite literal syntax with colon-separated key-value pairs, so it's easy to build them during initialization.

```
var timeZone = map[string]int{
    "UTC" : 0 + 60 * 60,
    "EST" : -5 + 60 * 60,
    ...
}
```

- Missing keys

An attempt to fetch a map value with a key that is not present in the map will return the zero value for the type of the entries in the map.

- Deleting Map Entries-:

To delete a map entry, use built-in delete function, whose arguments are the map and the key to be deleted. It's safe to do this even if the key is already absent from the map.

```
delete(timeZone, "PDT")
```

* Nested-:

Maps can contain maps, creating a nested structure for example.

```
map[string]map[string]int
map[bool]map[string]int
```